

TEST PLAN

1. Objective

To ensure that the Fliply application meets its specified requirements, performs reliably under expected and edge-case conditions, and provides a seamless and secure user experience. The test plan covers functional and non-functional aspects to guarantee software quality before release.

2. Resources

Personnel:

- QA/Test Engineer(s): Ngoc Nguyen, Thanh Nguyen, Nhut Vo, Hoang Vu
- Developers (for bug-fixing and clarifications): Ngoc Nguyen, Thanh Nguyen, Nhut Vo, Hoang Vu
- Project Manager/Product Owner: Teacher

Tools:

- Automated testing frameworks (Jupiter)
- Code coverage tools (Jacoco)
- Static code analysis (SonarQube)

Documentation:

- Project requirements specifications
- User stories/design documents
- Source code
- UML Diagrams

3. Test Environment

- Operating Systems: Windows 10/11, MacOS
- Devices: Desktop
- Database: MariaDB
- Test Data: Seeded test database with both valid and invalid inputs, edge cases, and empty states

4. Test Tasks

4.1 Functional Test Tasks

- Unit Testing
 - Test individual modules, functions, and methods for correctness.
 - Ensure all business logic is covered.
- Integration Testing
 - Verify interactions between components/modules.
 - Check data flow between layers.
- System/End-to-End Testing
 - Validate the system as a whole against requirements.
 - Scenario-based user journeys (happy path and alternate flows).
- User Interface Testing
 - Ensure UI elements function as intended (buttons, forms, navigation).

4.2 Non-functional Tasks

- Overview of performance testing results and any optimizations made.
- Heuristic evaluation report
- User Accepting test (UAT) results report
- Summary of security testing findings and actions taken to mitigate vulnerabilities.
- Description of UI/UX improvements and code optimizations.

5. Test Cases

5.1 TC01 – Register

5.1.1 TC01-01 – Register (Valid Input)

Type: Functional

Precondition:

- User has access to the application
- User is not logged in

Steps:

1. Launch the application
2. Click **Register** button
3. Enter valid **Last Name**
4. Enter valid **First Name**
5. Enter valid **Email** (not previously registered)
6. Enter valid **Password**
7. Select **Are you a teacher? = Yes/No**
8. Check **Agree with Terms & Policy**
9. Click **Register**

Expected:

- User account is successfully created
- User is redirected to homepage

5.1.2 TC01-02 – Register (Missing Required Fields)

Type: Functional

Precondition: User is on Register page

Steps:

1. Launch the application
2. Click **Register** button
3. Leave one or more required fields empty
4. Click **Register**

Expected:

- Error message displayed: "Please fill in all fields."
- Registration is not successful

5.1.3 TC01-03 – Register (Invalid Email Format)

Type: Functional

Precondition: User is on Register page

Steps:

1. Launch the application
2. Click **Register** button
3. Enter invalid email format (e.g., **abc123**)
4. Fill other fields correctly
5. Click **Register**

Expected:

- Error message for invalid email is displayed: "Email is not valid"
- Registration fails

5.1.4 TC01-04 – Register (Password less than 6 characters)

Type: Functional

Precondition: User is on Register page

Steps:

1. Launch the application
2. Click **Register** button
3. Enter valid **Last Name**
4. Enter valid **First Name**
5. Enter valid **Email**
6. Enter **Password with less than 6 characters** (e.g., 12345)
7. Select **Are you a teacher? = Yes/No**
8. Check **Agree with Terms & Policy**
9. Click **Register**

Expected:

- Error message displayed: "Password must be at least 6 characters"
- Registration is not successful

5.1.5 TC01-05 – Register (Email Already Exists)

Type: Functional

Precondition: Email is already registered in the system

Steps:

1. Launch the application
2. Click **Register** button
3. Enter valid **Last Name**
4. Enter valid **First Name**
5. Enter an **Email that already exists**
6. Enter valid **Password**
7. Select **Are you a teacher? = Yes/No**
8. Check **Agree with Terms & Policy**
9. Click **Register**

Expected:

- Error message displayed: "Email already exists"
- Registration is not successful

5.1.4 TC01-06 – Register (Terms Not Accepted)

Type: Functional

Precondition: User is on Register page

Steps:

1. Launch the application

2. Click **Register** button
3. Fill all fields correctly
4. Do NOT check **Agree with Terms & Policy**
5. Click **Register**

Expected:

- Error message displayed: "Please accept the terms and conditions"
- Registration is not allowed

5.2 TC02 – Login

5.2.1 TC02-01 – Login (Valid Credentials)

Type: Functional

Precondition: User account already exists

Steps:

1. Launch the application
2. Click **Login** button
3. Enter valid **Email**
4. Enter valid **Password**
5. Click **Login**

Expected:

- User is logged in successfully
- Redirected to homepage

5.2.2 TC02-02 – Login (Invalid Credentials)

Type: Functional

Precondition: User is on Login page

Steps:

1. Launch the application
2. Click **Login** button
3. Enter incorrect **Email or Password**
4. Click **Login**

Expected:

- Error message displayed: "Invalid email or password"
- Login fail

5.3 TC03 – Teacher Create Class

5.3.1 TC03-01 – Teacher Create Class (Valid)

Type: Functional

Precondition: User is logged in as **Teacher**

Steps:

1. Launch the application
2. Login as Teacher
3. Navigate to **Classes** menu
4. Click **+ Add more class**
5. Enter valid **Class Code**
6. Click **Add**

Expected:

- Class is successfully created
- Class appears in teacher's class list

5.3.2 TC03-02 – Teacher Create Class (Empty Class Code)

Type: Functional

Precondition: User is logged in as Teacher

Steps:

1. Launch the application
2. Login as Teacher
3. Navigate to **Classes** menu
4. Click **+ Add more class**
5. Leave **Class Code** empty
6. Click **Add**

Expected:

- Validation error displayed
- Class is not created

5.3.3 TC03-03 – Non-Teacher Attempt to Create Class

Type: Functional

Precondition: User is logged in as **Student (Non-teacher)**

Steps:

1. Login as Student
2. Try to access **Create Class** feature

Expected:

- Feature is hidden

5.4 TC04 – Teacher Creates Flashcard Set

5.4.1 TC04-01 – Teacher Creates Flashcard Set (Valid)

Type: Functional

Precondition: User is logged in as **Teacher**

Steps:

1. Launch the application
2. Login as **Teacher**
3. Navigate to **Classes** menu
4. Click **Class name**
5. Click “**+ Add more Flashcard Set**”
6. Enter valid Subject
7. Upload file csv
8. Click “**Add**”

Expected:

- Flashcard Set is successfully created
- Flashcard Set appears in class's flashcards set list

5.4.2 TC04-02 – Teacher Creates Flashcard Set (Empty Fields)

Type: Functional

Precondition: User is logged in as **Teacher**

Steps:

1. Launch the application
2. Login as **Teacher**
3. Navigate to **Classes** menu
4. Click **Class name**
5. Click “**+ Add more Flashcard Set**”
6. Leave Subject
7. Click “**Add**”

Expected:

- Validation error displayed
- Flashcard Set is not created

5.5 TC05 – Student Creates Flashcard

5.5.1 TC05-01 – Student Creates Flashcard (Valid)

Type: Functional

Precondition: User is logged in as **Student**

Steps:

1. Launch the application
2. Login as **Student**
3. Navigate to **Flashcards** menu
4. Click “+”
5. Choose Subject, Enter valid Term and Definition
6. Click “**Save**”

Expected:

- Flashcard is successfully created
- Flashcard appears in the selected set

5.5.2 TC05-02 – Student Creates Flashcard (Empty Term or Definition)

Type: Functional

Precondition: User is logged in as **Student**

Steps:

1. Launch the application
2. Login as **Student**
3. Navigate to **Flashcards** menu
4. Click “+”
5. Leave Term or Definition empty
6. Click “**Save**”

Expected:

- Validation error displayed
- Flashcard is not created

5.6 TC06 – Student Views Flashcard

Type: Functional

Precondition: Student has access to a Flashcard Set

Steps:

1. Launch the application
2. Login as **Student**
3. Navigate to **Flashcards** menu
4. Click a **Flashcard**
5. Click to view flashcard side by side

Expected:

- Flashcards are displayed correctly
- Each card shows Term and Definition

5.7 TC07 – Student Creates Quiz**5.7.1 TC05-01 - Student Creates Quiz (Valid)**

Type: Functional

Precondition: User is logged in as **Student**;

Steps:

1. Launch the application
2. Login as **Student**
3. Navigate to **Quizzes** menu
4. Click “+”
5. Enter number of questions to create quizzes
6. Click Generate

Expected:

- Quiz is successfully created
- Quiz appears in My Quizzes list

5.7.2 TC05-02 - Student creates Quiz (Empty flashcard)

Type: Functional

Precondition: User is logged in as **Student**;

Steps:

1. Launch the application
2. Login as **Student**
3. Navigate to **Quizzes** menu
4. Click “+”

5. Enter number of questions to create quizzes
6. Click Generate

Expected:

- Validation error is displayed
- Quiz is not created
- User is informed that there is no flashcards

5.7.3 TC05-03 - Student creates Quiz (Empty input or Input more than flashcards)

Type: Functional

Precondition: User is logged in as **Student**;

Steps:

1. Launch the application
2. Login as **Student**
3. Navigate to **Quizzes** menu
4. Click “+”
5. Let the input empty or Enter more than flashcards number
6. Click Generate

Expected:

- Validation error is displayed
- Quiz is not created
- User is informed about the specific errors

5.8 TC08 - Student Edits Flashcard

5.8.1 TC05-01 - Student Edits Flashcard (Valid)

Type: Functional

Precondition: User is logged in as **Student**;

Steps:

1. Launch the application
2. Login as **Student**
3. Navigate to **Flashcards** menu
4. Choose flashcard
5. Click “...” on the top left and choose **Edit**
6. Input the information need edited
7. Click **Save**

Expected:

- Flashcard is updated successfully
- New information is displayed in flashcard list and detail view
- No duplicate or extra card is created after editing

5.8.2 TC05-02 - Student Edits Flashcard (Information missing : Subject, Term, Definition)

Type: Functional

Precondition: User is logged in as **Student**;

Steps:

1. Launch the application
2. Login as **Student**
3. Navigate to **Flashcards** menu
4. Choose flashcard
5. Click “...” on the top left and choose **Edit**
6. Empty information edited
7. Click **Save**

Expected:

- Validation error is displayed
- Flashcard is not updated
- Empty Term or Definition is rejected

5.9 TC09 - Student Deletes Flashcard

5.9.1 TC05-01 - Student Deletes Flashcard (Valid)

Type: Functional

Precondition: User is logged in as **Student**;

Steps:

1. Launch the application
2. Login as **Student**
3. Navigate to **Flashcards** menu
4. Choose flashcard
5. Click “...” on the top left and choose **Delete**
6. Click

Expected:

- Flashcard is deleted successfully
- Flashcard no longer appears in the set

- Flashcard count is updated correctly

5.10 TC10 - Student Studies Quizzes

5.10.1 TC10-01 - Student Studies Quiz (Valid)

Type: Functional

Precondition:

- User is logged in as Student
- At least one Quiz exists in My Quizzes.

Steps:

1. Launch the application
2. Login as Student
3. Navigate to Quizzes menu
4. Select a quiz from My Quizzes
5. Click Start Quiz
6. Answer the quiz questions
7. Click Submit / Finish

Expected:

- Quiz results are shown

5.10.2 TC10-02 - Student Studies Quiz (Empty)

Type: Functional

Precondition:

- User is logged in as Student
- Selected quiz is empty or unavailable.

Steps:

1. Launch the application
2. Login as Student
3. Navigate to Quizzes menu

Expected:

- There is no quiz on the screen
- Total quiz: 0

5.11 TC11 - User Chooses Language

5.11.1 TC11-02 - User Chooses Language (Welcome Screen)

Type: Functional

Precondition: User is on the welcome screen

Steps:

1. Launch the application
2. Click on "Language"
3. Choose a supported language
4. Confirm the selection

Expected:

- Application UI is updated to the selected language
- All labels, buttons, and messages are translated correctly
- No UI layout issues occur

5.11.2 TC11-02 - User Chooses Language (Setting Screen)

Type: Functional

Precondition: User is on the setting screen

Steps:

1. Launch the application -> Register or Login
2. Open Account Screen
3. Click on "Language"
4. Choose a supported language
5. Confirm the selection

Expected:

- Application UI is updated to the selected language
- All labels, buttons, and messages are translated correctly
- No UI layout issues occur

5.12 TC12 - User Reload language

Type: Functional

Precondition: User has selected a non-default language

Steps: Reload the page or refresh the application

Expected:

- Selected language remains unchanged after reload
- Application does not revert to the default language
- UI content is displayed correctly

6. Deliverables

- Test cases/scripts (manual and automated)
- Test summary reports and defect logs

- Code coverage and quality reports